P2.3.2

The result of a measurement of the voltage standing wave pattern on a transmission line with source impedance $Z_0 = 50\Omega$ and characteristic impedance $Z_0 = 50\Omega$ is illustrated in Figure P2.3.2.1.

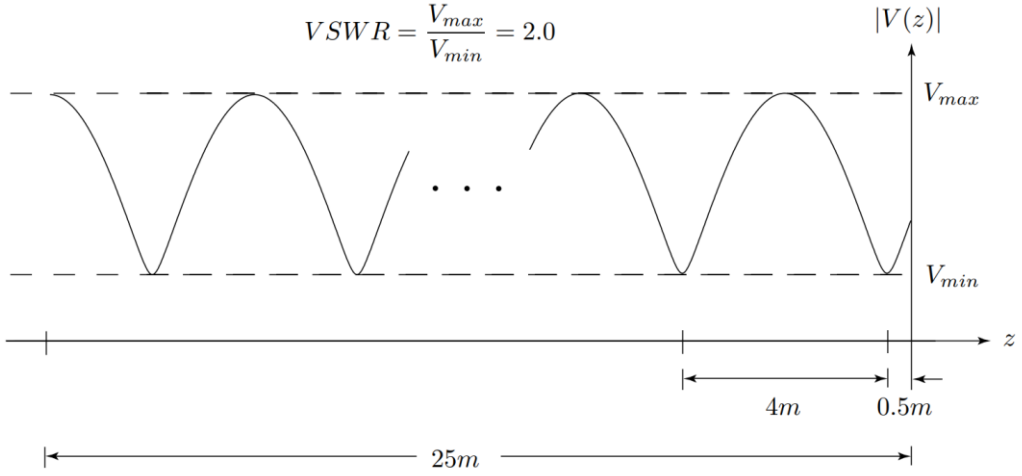


Figure P2.3.2.1 Voltage standing wave pattern.

(a) From the given data, determine the wavelength and load impedance Z_L.
(b) What is the input impedance at the source $Z(z = -25 \text{ m})$?
(c) Determine the complex load voltage $V_L = V(z = 0)$ in terms of V_s.
(d) What is the time-averaged power dissipated in the load Z_L?

P2.3.3

The "current standing wave pattern" of a TEM transmission line with characteristic impedance $Z_0 = 50\,\Omega$, permeability μ_0 and permittivity $\epsilon = 4\epsilon_0$ is shown in Figure P2.3.3.1.

(a) What is the frequency of excitation $f = \omega/2\pi$?
(b) Calculate the reflection coefficient Γ_L.

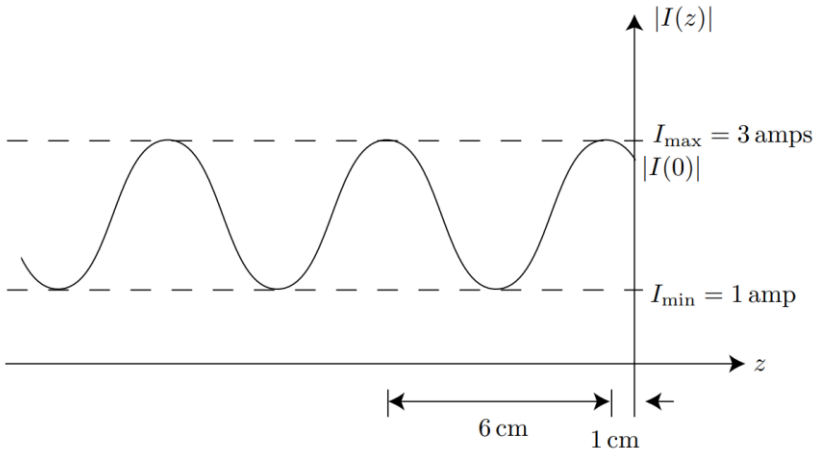


Figure P2.3.3.1

(c) Determine the load impedance Z_L in Ω.
(d) What is the time-average power flow along the line? Give a numerical answer.